MIL-S-43877

11 March 1974

MILITARY SPECIFICATION

SPINACH, DEHYDRATED, COMPRESSED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers dehydrated compressed spinach for use by the Armed Forces as an item of general issue.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-495	-	Coatings Exterior for Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-29	-	Canned Subsistence Items, Packaging and Packing Of
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter

MILITARY

MIL-L-1497	-	Labeling of Metal Cans for Subsistence Items
MIL-L-35078	-	Loads, Unit; Preparation of Nonperishable Subsistence

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-668	-	Sanitary Standards for Food Plants

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8915

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PUBLICATIONS

U.S. Department of Health, Education, and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20250.)

U.S. Department of Agriculture

United States Standards for Condition of Food Containers

(Application for copies should be addressed to the Information Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, Box 540, Benjamin Franklin Station, Washington, DC 20044.)

National Research Council

Food Chemicals Codex

(Application for copies should be addressed to the Office of the Director, National Academy of Sciences, National Research Council, Washington, DC 20418.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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American Dry Milk Institute, Inc.

Standards for Grades of Dry Milks, Including Methods of Analysis: Bulletin 916

(Application for copies should be addressed to the American Dry Milk Institute, Inc., 130 North Franklin Street, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Bid sample approval. Twelve, unless otherwise specified (see 6.1), hermetically sealed cans of at least 6 avoirdupois ounces each, representative of the product the supplier proposes to furnish, shall be packaged under nitrogen gas and submitted to the contracting officer prior to the bid opening. The contracting officer will have the samples tested to determine whether the product has the desired subjective quality characteristics specified in 3.4. If the bid is accepted, the remaining samples shall be held at a temperature of 40°F. or lower and shall be used as a reference standard for determining compliance of deliveries with subjective quality requirements. Additional samples may be withdrawn from accepted lots of materials for use as reference standards as deemed necessary by the inspecting agency. The contracting officer's approval of the bid sample does not relieve the supplier from complying with all other requirements of the specification not stipulated above.

3.2 Material.

3.2.1 Spinach. The spinach shall be of a clean sound freezer variety.

3.2.2 Sodium sulfite or sodium bisulfite. Sodium sulfite or sodium bisulfite shall conform to the requirements of the Food Chemicals Codex.

3.3 Processing.

3.3.1 Preparation. The product shall be prepared from fresh, sound, succulent leaves and adjoining stems which have been sorted, trimmed and cleaned by proper washing to remove sand, dirt, silt or any other undesirable substance. The spinach shall be cooked until tender.

3.3.2 Sulfiting. The spinach shall receive a sulfite treatment by such methods as will insure uniform distribution throughout the product, and a residual sulfur dioxide content in the dehydrated product within the limitations specified in 3.4.1. Contact of the wet spinach with iron, subsequent to the sulfiting operation, shall be strictly avoided.

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3.3.3 Freezing. The sulfited spinach shall be frozen as rapidly as possible. After freezing the product shall not be stored at temperatures above 0°F.

3.3.4 Dehydration. The product shall be freeze dehydrated (i.e., conversion of the water directly from solid to vapor phase, omitting the liquid phase entirely) at an absolute pressure not to exceed 1.0 millimeter of mercury. The product shall not be allowed to thaw prior to dehydration. After dehydration is completed, the pressure shall be equalized to atmospheric level with nitrogen. The product shall be conditioned, compressed and dried as specified in 3.3.5, 3.3.6, 3.3.7 and 3.4.1.

3.3.5 Conditioning. The freeze dried spinach shall be conditioned by adding moisture sufficient to make the product pliable so that it will not be crushed or ground under high pressure. (Note: Satisfactory compressed spinach has been produced by adding 8-14 percent moisture either by applying wet live steam for approximately five minutes or spraying evenly with a fine mist of water to increase the weight by 8-14 percent, placing the sprayed spinach in a closed container, and then heating (approximately 10 minutes at 150° F.), in order to equilibrate the moisture in the spinach.

3.3.6 Mechanical pressing. The moistened spinach shall be compressed into discs by applying approximately 1000 pounds per square inch of pressure for a dwell time of 10-20 seconds depending upon the variety and maturity of the spinach.

3.3.7 Redrying. The compressed spinach shall be redried within a minimum of time so that the finished product complies with 3.4.

3.4 Finished product.

3.4.1 Compressed product. The compressed discs shall be 3-3/4 ± 1/8 inches in diameter. The moisture content of the compressed spinach shall not exceed 2.5 percent by weight, determined as specified in 4.4.2.1, and the sulfur dioxide content shall be 400 ppm ± 200 ppm determined as specified in 4.4.2.2.

3.4.2 Rehydrated product. The compressed spinach shall rehydrate rapidly (less than 10 minutes) upon the addition of 16 parts of hot water (180-200° F.) to 1 part of spinach, by weight. The rehydrated product shall possess good color, flavor, odor and texture typical of freshly cooked spinach and shall be practically free from grit or sand.

3.5 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act, and regulations promulgated thereunder.

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3.6 Workmanship. The product shall be prepared, processed and packaged under modern sanitary conditions and in accordance with good commercial practice. The product shall be processed in establishments meeting the requirements of MIL-STD-668.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Pre-acceptance inspection. (Plant sanitation inspection). Product shall be rejected if produced in plants not meeting the requirements of 3.6.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.3.1.1 Sulfite. Conformance of sulfite to Food Chemicals Codex quality shall be ascertained by examination of labels on sulfite containers. Use of nonconforming sulfite shall be cause for rejection of the involved finished product.

4.3.1.2 Testing of cans. Can shall be tested for tin plate in accordance with 4.4.2.4. The sample unit shall be 1 can with lid. The sample size shall be the number of sample units indicated by inspection level S-1. Lot size shall be expressed in terms of cans. Test requirement for tin plate shall be an average requirement and results shall be reported to nearest 0.01 lb/base box. Testing of can label and external can coating (when required) shall be performed on the same cans submitted for tin plate test. A test failure shall indicate unacceptable cans and use of such shall be cause for rejection of the involved end item.

4.3.2 Process examination. Examination shall be performed to the extent necessary to determine conformance to processing requirements of 3.3. Nonconformance shall be cause for rejection of the involved product.

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4.3.3 Examination of end item. Examinations shall be in accordance with tables II and III and 4.3.3.1 through 4.3.3.3. Sampling plans for the aforementioned tables shall be in accordance with table I. The acceptable quality levels (AQLs) shall be expressed as defects per hundred units, except for table II which shall be expressed as percent defective.

TABLE I. Sampling plans

Table	Sample unit	Lot size expressed in	Inspection level	AQLs Major	Minor	Total
II	1 filled can	Cans	S-3	-	2.5	-
III	1 filled can	Cans	S-2	1.5	-	6.5

TABLE II. Examination of net weights 1/

Category	Defect
Minor	
201	More than 2/8 ounces under labeled net weight

1/ Weighing device shall be calibrated in increments no larger than 1/8 ounce. Lot shall be rejected if sample data indicate a lot average net weight less than labeled net weight or if label states less than 16 ounces.

TABLE III. Examination of product characteristics 1/

Category		Defect
Major	Minor	Dehydrated 2/
	201	Dimension less than 3-5/8 inches or more than 3-7/8 inches in diameter 3/
		Rehydrated 4/
101		Does not completely rehydrate within 10 minutes after adding hot water
102		Color not similar to bid sample

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TABLE III. Examination of product characteristics 1/ (cont'd)

Category		Defect
Major	Minor	Rehydrated 4/ (cont'd)
103		Flavor or odor not similar to bid sample
104		Texture not similar to bid sample
105		More than a trace of sand or grit as evidenced by mouth feel

1/ Presence of foreign material (e.g. insects, filth, glass, wood, metal, paint, etc.) shall be cause for rejection of the lot.

2/ Examine one disc from a sample unit.

3/ Measuring device shall be calibrated in no larger than 1/8 inch increments.

4/ Rehydrate one disc from a sample unit in accordance with 3.4.2. Drain excess water.

4.3.3.1 Examinations of cans. Examination of filled and sealed cans for external condition and labeling shall be in accordance with US Standards for Condition of Food Containers. Examination of cans for seam leakage shall be in accordance with the quality assurance provisions of PPP-C-29.

4.3.3.2 Examination of shipping containers. The filled and closed shipping containers shall be examined in accordance with the quality assurance provisions in the appendix of PPP-B-636. In addition, the following defects shall be included in the examination. Major - Markings missing, incorrect or illegible; Minor - Number or arrangement of contents not as required. For containers required to be in accordance with 5.2.3.1, only the aforementioned defects pertinent to marking, and number and arrangement of contents shall apply.

4.3.3.3 Examination of unit loads. Unit loads shall be examined in accordance with the quality assurance provisions of MIL-L-35078.

4.3.4 Testing oxygen content of cans and testing of finished product. The filled cans shall be tested for oxygen content, and the finished product shall be tested for moisture and sulfur dioxide content. Test procedures shall be in accordance with 4.4. The sample unit for testing oxygen content shall be

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1 filled and sealed can and the sample size shall be the number of cans indicated by inspection level S-2. Lot size shall be expressed in cans. The acceptable quality level expressed as percent defective shall be 2.5. A test result of more than 2.0 percent (to nearest 0.1 percent) shall be scored as a defect. The sample for testing moisture and sulfur dioxide content shall be a 4-ounce composite of product derived from the number of cans indicated by inspection level S-2. Results shall be reported to the nearest 0.1 percent for moisture and nearest ppm for sulfur dioxide. A test failure shall be cause for rejection of the lot.

4.4 Test methods. Examination and test procedures which differ from those specified herein, unless otherwise excepted, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum such quality assurance, the supplier will use the procedures set forth herein. In case of dispute as to examination or test results, the procedures specified herein will govern.

4.4.1 Examination procedures.

4.4.1.1 Leakage. Procedure for determining leakage on can seams shall be in accordance with PPP-C-29.

4.4.2 Test procedures.

4.4.2.1 Moisture. Moisture determination shall be made by the vacuum oven method as specified in 4.4.2.1.1 and 4.4.2.1.2. (NOTE: Samples for moisture determination shall be collected in clean, dry, jars which shall be completely filled and sealed with air-tight lids.)

4.4.2.1.1 Preparation of sample. Prepare a clean, dry Waring blender or similar mill by first grinding a portion of the sample and then discarding it. Grind approximately 100 grams of the sample for one minute. Place the ground material in a small, dry tightly-closed container for at least one-half hour. Pass the ground material over a U.S. Standard No. 20 sieve (0.0331-inch openings). That portion which passes through the U.S. Standard No. 20 sieve is to be analyzed for moisture, and shall be weighed immediately or placed in a small, dry, container with a tightly-closed lid until ready for analysis (a one-half pint Mason jar is satisfactory).

4.4.2.1.2 Moisture determination. Moisture determination procedure shall be as follows: Weigh two, approximately 3-gram samples of the ground screened material (that portion which has passed through the U.S. Standard No. 20 sieve) into tared, dry, aluminum weighing dishes (approximately 2 or 2-1/2 inch in

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diameter and 3/4 inch in depth) with tight-fitting covers. Place the dishes with cocked lids in a vacuum oven for 6 hours at a temperature of 70° C, under a pressure of not more than 100 mm of mercury. Remove the dishes and allow them to cool in a desiccator before weighing. Calculate the percent of moisture in the original samples.

4.4.2.2 Sulfur dioxide content. Sulfur dioxide content of the ground sample shall be determined in accordance with Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Preservatives and Artificial Sweeteners, Section: Total Sulfurous Acid.

4.4.2.3 Oxygen content. Oxygen content shall be determined in accordance with American Dry Milk Institute Inc. publication, Bulletin 916, Standards for Grades of Dry Milks: Chapter: Methods of Analysis: Determination of Oxygen. When the can has been sealed under reduced pressure, the gases may have to be extracted and compressed prior to testing for oxygen.

4.4.2.4 Tin coating weight. Tin coating weight shall be determined by any method specified in PPP-C-96.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C, as specified (see 6.1).

5.1.1 Level A. Sufficient discs to fill the containers, having a net weight of 16 ounces or more shall be packaged in a size 401 by 411, open top style, round, metal can with soldered side seam and compound-lined, double-seamed ends. Individual cans may vary no more than 2/8 ounces from the labeled net weight, provided the lot average net weight is not less than the labeled net weight. The can shall be made throughout from not less than 0.25-pound electrolytic tin plate per base box (see 4.4.2.4) and shall be coated overall on the outside with a coating conforming to type I of TT-C-495. The product shall be packaged under an atmosphere of nitrogen so that the oxygen content of the gases in the filled and sealed container shall not exceed 2.0 percent when tested in accordance with 4.4.2.3. The can shall not leak when examined in accordance with 4.4.1.

5.1.1.1 Level C. The product shall be packaged in accordance with 5.1.1 except that cans with or without commercial exterior coating will be acceptable.

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5.2 Packing. Twenty-four cans of the product, arranged 4 in length, 3 in width and 2 in depth, shall be packed on end in a snug-fitting shipping container in accordance with level A, B, or C, as specified (see 6.1). The RSC box may be top or end opening. Strapping of the end opening box shall be in the same relation to the contents as for the top opening box.

5.2.1 Level A. The shipping container shall be a fiberboard box with sleeve, constructed, closed, and reinforced in accordance with style RSC-SL, V2s, of PPP-B-636. Sleeve and reinforcement are not required when unit loading is specified.

5.2.2 Level B. The shipping container shall be a fiberboard box, constructed, closed, and reinforced in accordance with style RSC, W5c or W5s, of PPP-B-636.

5.2.2.1 When specified (see 6.1 and 6.3), the shipping container specified in 5.2.2 shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636, except that two reinforcing bands may be used, one lengthwise and one girthwise.

5.2.3 Level C. The shipping container shall be a fiberboard box, constructed and closed in accordance with style RSC, type CF or SF, class domestic, method II closure, of PPP-B-636.

5.2.3.1 When specified (see 6.1 and 6.4), the shipping container shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified.

5.4 Labeling and marking.

5.4.1 Cans. The cans shall be labeled in accordance with MIL-L-1497. Labeling shall include directions for use and the statement:

GAS PACKED

PRESERVED WITH SULFUR DIOXIDE

DIRECTIONS FOR USE

Place contents of can (16 oz) in serving pan. Add 6-3/4 quarts of boiling salted

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water (2-1/2 tbsp salt). Cover and let stand 10 minutes before serving. Stir after 5 minutes to break apart. Season as desired. Makes 54 servings (1/3 cup each). (NOTE: Use 1 pound dehydrated spinach for 10 pounds frozen spinach.)

5.4.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

5.4.3 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When continuous inspection of processing is required (see 4.3.2).
(c) Level of packaging and packing required (see 5.1 and 5.2).
(d) When packing specified in 5.2.2.1 or 5.2.3.1 is required.
(e) Type and class of unit load when unit loading is specified (see 5.3).

6.2 Based on conditions known or expected to be encountered during shipping, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15, NAVSUPINST 4030.28, AFR 71-6, MCO 4030.14D and DSAR 4145.7, as applicable.

6.3 Packing specified in 5.2.2.1 is intended for transfer at sea operations or specific overseas operations.

6.4 Packing specified in 5.2.3.1 is intended for direct shipment from the supply source to the first receiving activity for immediate use within CONUS.

Custodians:

Army - GL
Navy - SA
Air Force - 45

Preparing activity:

Army - GL

Project No. 8915-0460

Review activities:

Army - MD
Navy - MC, MS
DP - SS

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DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-S-43877 SPINACH, DEHYDRATED, COMPRESSED

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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